

Block Diagrams and Mason's Gain Formula

Reducing an interconnection of blocks to one transfer function.

Ogata, *Modern Control Engineering*, Ch. 2.4–2.5.

In this tutorial you will:

- apply the series, parallel, and feedback reduction rules,
- use **series**, **parallel**, and **feedback** to combine blocks, and
- cross-check a multi-loop diagram with Mason's gain formula.

Step through with **Ctrl+Enter**, or render a report with **publish**.

Complex control systems are described by interconnected blocks, each representing a transfer function, joined by summing junctions and takeoff points. Three reduction rules let any block diagram be collapsed to a single transfer function:

- **Series (cascade):** blocks $G_1(s)$ and $G_2(s)$ in series combine to $G_1(s)G_2(s)$.
- **Parallel:** blocks $G_1(s)$ and $G_2(s)$ summed at a junction combine to $G_1(s) \pm G_2(s)$.
- **Feedback:** a forward path $G(s)$ with feedback path $H(s)$ closes to

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$$

with $-$ for negative feedback and $+$ for positive feedback.

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Series and Parallel Combination

```
G1 = tf(1,[1 1]);    % 1/(s+1)
G2 = tf(2,[1 2]);    % 2/(s+2)
```

```
G_series = series(G1,G2)
G_parallel = parallel(G1,G2)
```

```
G_series =
```

$$\frac{2}{s^2 + 3s + 2}$$

```
Continuous-time transfer function.
```

```
G_parallel =
```

$$\frac{3s + 4}{s^2 + 3s + 2}$$

```
Continuous-time transfer function.
```

Closing a Negative-Feedback Loop

For unity feedback ($H(s) = 1$):

```
G = tf(10,[1 2 0]); % 10/(s(s+2))
T_unity = feedback(G,1)
```

T_unity =

$$\frac{10}{s^2 + 2s + 10}$$

Continuous-time transfer function.

For non-unity feedback $H(s) = 1/(s + 5)$:

```
H = tf(1,[1 5]);
T = feedback(G,H)
```

T =

$$\frac{10s + 50}{s^3 + 7s^2 + 10s + 10}$$

Continuous-time transfer function.

Mason's Gain Formula

For diagrams too complex for simple reduction, Mason's gain formula gives the overall transfer function directly from the signal-flow graph:

$$T = \frac{Y}{R} = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

where P_k is the gain of the k -th forward path, Δ is the determinant of the graph,

$$\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$$

(L_i = individual loop gains, sums taken over non-touching loops), and Δ_k is Δ evaluated on the subgraph that does not touch forward path k .

Worked Example: Two-Loop System

Forward path: $P_1 = G_1 G_2 G_3$. Loops: $L_1 = -G_1 G_2 H_1$ (around $G_1 G_2$), $L_2 = -G_2 G_3 H_2$ (around $G_2 G_3$). L_1 and L_2 share G_2 , so they touch and there is no second-order term.

$$\Delta = 1 - (L_1 + L_2) = 1 + G_1 G_2 H_1 + G_2 G_3 H_2$$

Since the forward path touches every loop, $\Delta_1 = 1$, giving

$$T = \frac{G_1 G_2 G_3}{1 + G_1 G_2 H_1 + G_2 G_3 H_2}$$

Verify symbolically with $G_1 = 1/(s + 1)$, $G_2 = 1/s$, $G_3 = 2$, $H_1 = 1$, $H_2 = 0.5$.

```
syms s
G1s = 1/(s+1);
G2s = 1/s;
G3s = 2;
H1s = 1;
H2s = sym(1)/2;
```

```
P1 = G1s*G2s*G3s;
Delta = 1 + G1s*G2s*H1s + G2s*G3s*H2s;
T_mason = simplify(P1/Delta)
```

```
T_mason =

2/(s^2 + 2*s + 2)
```

Cross-check by building the same diagram with `series`, `parallel`, and `feedback` using numeric transfer functions: inner loop G_1G_2 closed by H_1 , then cascaded with G_3 and closed by H_2 around the outer loop.

```
G1n = tf(1,[1 1]);
G2n = tf(1,[1 0]);
G3n = tf(2,1);
H1n = tf(1,1);
H2n = tf(0.5,1);

inner = feedback(series(G1n,G2n),H1n);
T_blocks = feedback(series(inner,G3n),H2n)
```

```
T_blocks =

      2
-----
s^2 + s + 2
```

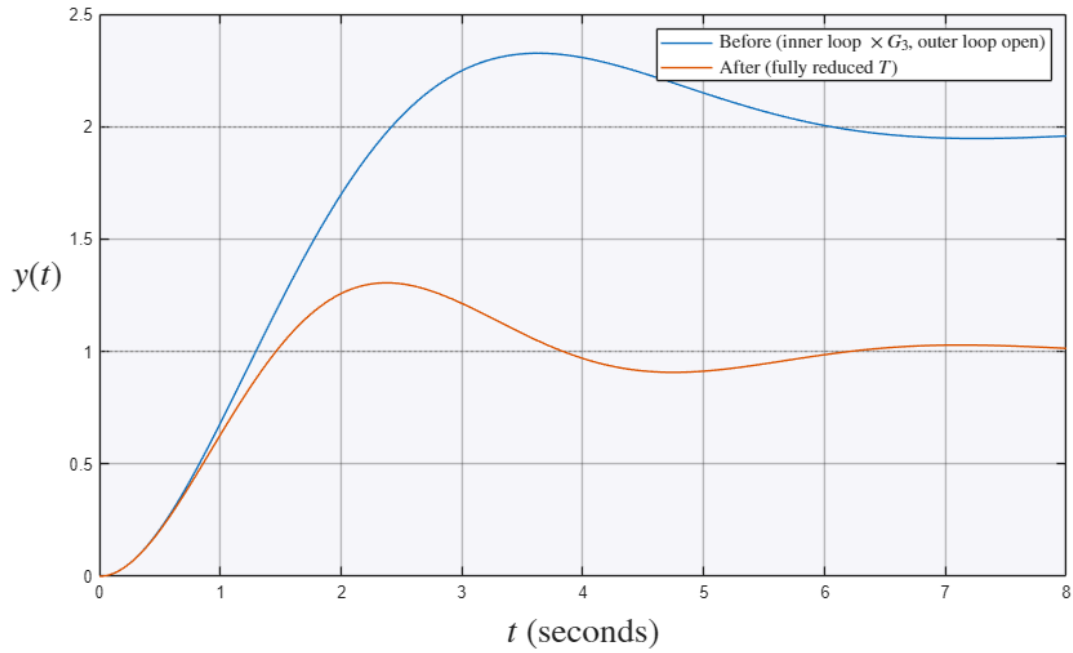
Continuous-time transfer function.

Before vs. After Reduction: The Inner Loop vs. the Whole Diagram

Block-diagram algebra collapses the multi-block diagram step by step. Compare the inner loop alone (before the outer feedback is closed) with the fully reduced system `T_blocks` (after) – closing the outer loop speeds up and re-damps the response.

```
figure
step(series(inner,G3n),0:0.01:8)
hold on
step(T_blocks,0:0.01:8)
hold off
grid on
legend('Before (inner loop  $\times G_3$ , outer loop open)', 'After (fully reduced  $T$ )', ...
      'Interpreter','latex','FontSize',11)
title('Block Reduction: Before vs. After Closing the Outer Loop','Interpreter','latex','FontSize',15)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

Block Reduction: Before vs. After Closing the Outer Loop



Try it yourself

- Flip the inner loop to positive feedback with `feedback(series(G1n,G2n),H1n,+1)` and notice the poles move the other way.
- Set $H2s = 1$ and re-derive T_{mason} , then check it against the numeric `feedback` result – they should still agree.